

Assignment 4: Rubric Mapping

Smart Water Capstone

Mahmudul Hasan (4125999049)
Xi'an Jiaotong University · 2026

1 Technical Implementation (30%)

Criterion	Evidence
Code runs	<code>streamlit run app/main.py</code> ; <code>pytest -q</code> → 33 passed
Architecture	<code>docs/ARCHITECTURE.md</code> ; pipeline figure in README
Correct outputs	Four modules: weather, runoff, reservoir, flood

2 AI Collaboration Log (30%)

Criterion	Evidence
AGENTS.md	Root AGENTS.md + per-experiment copies
Prompt log	<code>prompt.log.md</code>
Corrections	9 documented in <code>assignment4/AI_FAILURES.md</code>
Jagged Frontier	<code>docs/JAGGED_FRONTIER.md</code>
Case study	<code>submission/portfolio/AI_Engineering_Portfolio.pdf</code>

3 Physical Validity (20%)

Rule	Evidence
$Q \leq P$	<code>validation.py</code> , <code>test_runoff.py</code>
Storage bounds	<code>validate_storage_mcm</code>
Monotonic flood	Exp 4 validation 9/9 PASS
Reference case	$Q = 13.80$ mm at $P = 50$, CN=80

4 Presentation (20%)

Item	Evidence
5-min outline	<code>docs/PRESENTATION_OUTLINE.md</code>
Demo script	<code>docs/DEMO_SCRIPT.md</code>
Tool evaluation	JAGGED_FRONTIER tool comparison
Live demo	Student presentation (rehearsal required)

5 Repository

<https://github.com/mahmud456alhasan-debug/smart-water-capstone>

6 Assignment 3 link

Swiss Cheese package: `assignment3/` (33 tests + hallucination case study).